

INF01090 – Data Science

Final Course Project

Semester 2026/1

Objective

The final project aims to integrate the main concepts studied throughout the course into a realistic Data Science problem. Each group must choose an interesting problem from any application domain (healthcare, sports, economics, social networks, environment, games, music, etc.) and develop a solution following the stages of the Data Science pipeline.

The project should demonstrate:

- the ability to formulate a relevant problem;
- understanding of the Data Science pipeline;
- mastery of the techniques studied in the course;
- critical analysis and communication of results.

Project Format

Groups may have up to **4 students**.

The project must include:

1. a clear definition of the problem;
2. dataset acquisition and description;
3. exploratory data analysis;
4. data cleaning and transformation;
5. feature engineering (when applicable);
6. training of at least one:
 - regression model; or
 - classification model.
7. quantitative evaluation of the model;
8. interpretation and critical discussion of the results.

Expected Pipeline

Whenever applicable, the project should address the following stages:

- problem definition;
- data collection/selection;
- data cleaning;
- data transformation;
- visualization and exploratory analysis;
- modeling;
- evaluation;
- interpretation of results;
- limitations and future work.

Dataset

The dataset may be:

- publicly available;
- obtained through APIs;
- collected by the group;
- taken from Kaggle competitions;
- obtained from scientific repositories.

The chosen dataset should be sufficiently interesting to support meaningful analysis and predictive modeling.

Deliverables

Each group must submit:

1. a Jupyter Notebook or Google Colab notebook;
2. the source code used in the project;
3. the dataset (or clear instructions for obtaining it);
4. a short report describing:
 - objective;
 - dataset;
 - methodology;
 - results;
 - conclusions.

The notebook must be executable on another machine.

Final Presentation

Presentations will take place in two sessions:

- **June 23:** Regular presentation session.
- **June 25:** Project showcase and competition session.

Each group must give a final presentation in a *pitch* format with a maximum duration of:

10 minutes

The presentation should be concise, clear, and focused on the main aspects of the project. Groups presenting on June 25 will participate in a friendly project showcase competition evaluated by a committee. The committee will select the **Top 5 Projects** considering aspects such as originality, technical quality, clarity of presentation, and overall impact. The two sessions are designed to accommodate different presentation preferences and project styles while ensuring that all groups have the opportunity to present their work in a supportive environment. Students are encouraged to select the session that best aligns with their project goals, presentation style, and personal comfort level.

Participation in the competition session is completely optional. The choice of presentation day does not affect the final course grade.

Suggested Pitch Structure

1. problem and motivation;
2. dataset used;
3. developed pipeline;
4. main visualizations;
5. model(s) used;
6. metrics and results;
7. final demonstration;
8. conclusions.

Evaluation Criteria

Projects will be evaluated considering:

- relevance and clarity of the problem;
- quality of the exploratory analysis;
- quality of data cleaning and transformation;
- adequacy of the modeling approach;
- interpretation of results;
- quality of visualizations;
- organization of the notebook and code;
- quality of the presentation;
- final demonstration.

Good luck and be creative!